

Author: Don't you feel that the phrase "velocity at a given moment of time" is merely a paraphrase of the "instantaneous velocity"? Six of one and half a dozen of the other, eh? The term "velocity at a given moment of time" calls for an explanation as much as the term "instantaneous velocity".

To measure the velocity of a body, one should obviously measure a certain distance (path) covered by the body, and the time interval during which the distance is covered. But, what path and period of time are meant when we refer to the velocity at a given moment of time?

Reader: Yes, in order to measure velocity, one must actually know a certain path and time interval during which the path is covered. But our subject is not the measurement, it is a definition of the instantaneous velocity.

Author: For the time being we shall not bother about a formal definition. It is more ~~and~~ important to realize its essential meaning. In order to do this, we cannot ~~and~~ avoid the aspect of measurements. Now, how would you find a way to measure the velocity of a body at a given moment of time?

Reader: I can take a short time interval, Δt , that is, the period from the given moment of time t to the moment $t + \Delta t$. During this time interval the body covers a distance Δs . If Δt is sufficiently small, the ratio $\frac{\Delta s}{\Delta t}$ will give the velocity of the body at the moment t .

Author: What do you mean by a sufficiently short time interval? What do you compare it with? Is this interval sufficiently small in comparison with a year, a month, an hour, a minute, a second, or a millisecond?

Reader: Perhaps, neither a year, a month, an hour nor a minute will do in this case. I see now that the instantaneous velocity can only be measured with a certain degree of accuracy. The smaller is Δt the smaller is the error with which the instantaneous velocity is measured.